

12.432

EE25BTECH11002 - Achat Parth Kalpesh

Question:

If any two columns of $\mathbf{P} = \begin{pmatrix} 4 & 7 & 8 \\ 3 & 1 & 5 \\ 9 & 6 & 2 \end{pmatrix}$ are interchanged, which one of the following statements regarding the value of the determinant is CORRECT?

- 1) Absolute value remains unchanged but sign will change.
- 2) Both absolute value and sign will change.
- 3) Absolute value will change but sign will not change.
- 4) Both absolute value and sign will remain unchanged.

Solution:

$$\mathbf{P} = \begin{pmatrix} 4 & 7 & 8 \\ 3 & 1 & 5 \\ 9 & 6 & 2 \end{pmatrix} \quad (4.1)$$

Interchanging columns 1 and 2 of $\mathbf{P}$ is equivalent to multiplying $\mathbf{P}$ on the right by the elementary matrix $\mathbf{E}$ that interchanges columns 1 and 2:

$$\mathbf{E} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (4.2)$$

Hence, the new matrix is

$$\mathbf{P}' = \mathbf{PE} \quad (4.3)$$

We know that

$$\det(\mathbf{P}') = \det(\mathbf{PE}) = \det(\mathbf{P}) \cdot \det(\mathbf{E}) \quad (4.4)$$

Now, the determinant of $\mathbf{E}$ is

$$\det(\mathbf{E}) = \begin{vmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix} = -1 \quad (4.5)$$

Hence,

$$\det(\mathbf{P}') = \det(\mathbf{P}) \cdot \det(\mathbf{E}) \quad (4.6)$$

$$= \det(\mathbf{P}) \cdot (-1) \quad (4.7)$$

$$= -\det(\mathbf{P}) \quad (4.8)$$

Hence, on interchanging any two columns of a matrix:

The sign of the determinant changes, but the absolute value remains the same